

Energy-Gated Frequency Networks

Henry

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Abstract. This work investigates whether low-pass, high-pass, and band-pass filtering can be embedded in a neural network so that energy in specific spectral regions modulates the activations used for audio classification. We introduce the Energy-Gated Frequency Network (EGFN), a raw-waveform architecture that exposes per-band filter responses, energy, and learned gates before temporal classification. The study progresses from controlled sinusoids to spoken digits, keyword spotting, and industrial valve anomaly detection. Free FIR kernels improve flexibility while losing their original physical band interpretation; Sinc-parameterized filters preserve explicit cutoffs but reduce performance in the controlled screening experiment. In a preliminary three-seed MIMII Valve comparison, a wide EGFN produced higher calibrated F1 than a smaller Conv1D baseline, although the architectures were not capacity matched. In a separate single-seed capacity-matched screening experiment, Conv1D achieved higher F1, while free-filter EGFN retained a small ROC–AUC advantage. These results are descriptive and do not establish statistical superiority. EGFN is therefore presented as a frequency-structured research architecture for studying spectral energy, adaptive modulation, and the trade-off between physical interpretability and predictive flexibility.

Keywords: raw audio, learnable filter bank, energy gating, interpretability, keyword spotting, industrial anomaly detection.

1 Introduction

Audio classifiers based on one-dimensional convolution can learn effective waveform representations directly from data. Their first-layer kernels can be interpreted as filters, but standard training does not require those kernels to remain low-pass, high-pass, or band-pass systems. This project began with a physical question: if classical filters separate sinusoidal components in electrical and digital signals, can the same principle organize neural activation for audio classification?

The central research question is:

Can low-pass, high-pass, and band-pass filtering be embedded within a neural network so that energy in specific spectral regions selectively controls the activations used for audio classification?

The goal is not to announce a universally superior architecture. The goal is to answer the question through progressively harder experiments, including negative and ambiguous results. The paper follows the concise progression from mechanism to controlled evaluation used in modern architecture papers such as Vaswani et al. [1], without implying any architectural relationship to the Transformer.

The main contributions are:

1. a differentiable frequency–energy–gate block for raw waveforms;
2. visual evidence of how band energy and gates respond to input frequency and task difficulty;

3. experiments across synthetic signals, FSDD, Speech Commands, and MIMII;
4. a three-seed industrial comparison followed by a capacity-matched control;
5. an explicit analysis of observability, physical interpretability, and parameter efficiency.

2 Background and Motivation

2.1 From neurons to waveform classifiers

A feed-forward neural network transforms an input $\mathbf{x}$ through hidden layers and produces output logits:

$$\mathbf{h}^{(\ell)} = \phi\left(W^{(\ell)}\mathbf{h}^{(\ell-1)} + \mathbf{b}^{(\ell)}\right), \quad (1)$$

$$\mathbf{o} = W^{(L)}\mathbf{h}^{(L-1)} + \mathbf{b}^{(L)}. \quad (2)$$

The input contains observations, hidden layers construct task-relevant representations, and the output scores are converted into class probabilities. Nonlinear activations prevent the entire stack from collapsing into a single linear transformation [2].

For mono audio, the input is a sampled sequence $x[n]$ represented as $\mathbf{x} \in \mathbb{R}^{B \times 1 \times T}$. A Conv1D layer applies learned finite kernels:

$$u_k[n] = \sum_{m=0}^{K-1} h_k[m]x[n-m]. \quad (3)$$

Thus Conv1D already performs learned filtering. The distinction in this work is not “filtering versus convolution,”

but whether frequency structure, energy, and modulation are explicitly parameterized and exposed.

2.2 Structured learnable audio frontends

SincNet learns only low and high cutoff frequencies of first-layer band-pass filters [3]. LEAF extends learnable audio frontends with filtering, pooling, compression, and normalization [4]. These works motivate a useful middle ground between fixed signal-processing features and unconstrained convolution. EGFN explores a related but distinct question: can explicit per-band energy drive a learned multiplicative gate, and what does that gate reveal about model behavior?

The Sinc mode evaluated here is a Sinc-parameterized frontend, not a complete reimplement of SincNet. Likewise, LEAF contains additional learnable pooling, compression, and normalization components. No direct SincNet or LEAF baseline is reported; this limits claims about novelty relative to established learnable frontends.

3 Energy-Gated Frequency Network

3.1 Filter-bank frontend

The frontend receives $\mathbf{x} \in \mathbb{R}^{B \times 1 \times T}$ and produces M aligned band responses. For an odd kernel length $K = 101$, a windowed low-pass response is initialized as

$$h_{\text{LP}}[n; f_c] = 2 \frac{f_c}{f_s} \text{sinc}\left(2 \frac{f_c}{f_s} n\right) w[n], \quad (4)$$

where $w[n]$ is a Hamming window for the fixed and free modes. High-pass and band-pass filters follow from spectral inversion and subtraction:

$$h_{\text{HP}}[n; f_c] = \delta[n] - h_{\text{LP}}[n; f_c], \quad (5)$$

$$h_{\text{BP}}[n; f_l, f_h] = h_{\text{LP}}[n; f_h] - h_{\text{LP}}[n; f_l]. \quad (6)$$

Three modes were evaluated:

- **Fixed:** initialized FIR coefficients remain constant.
- **Free:** every FIR coefficient becomes trainable.
- **Sinc:** only ordered low and high cutoffs are learned, while a Hann-windowed band-pass kernel is reconstructed during each forward pass.

The eight-band fine initialization is < 250 , 250 – 500 , 500 – 750 , 750 – 1000 , 1000 – 1500 , 1500 – 2500 , 2500 – 4000 , and > 4000 Hz. In free mode these names describe initialization only; they do not guarantee the learned spectral support.

3.2 Energy-dependent gating

After filtering, bias and GELU activation are applied:

$$u_k[n] = (h_k * x)[n], \quad (7)$$

$$a_k[n] = \text{GELU}(u_k[n] + b_k). \quad (8)$$

The mean-square energy of each unactivated response is

$$E_k = \frac{1}{T} \sum_{n=1}^T u_k[n]^2. \quad (9)$$

For independent gates,

$$g_k = \sigma(\alpha_k \log(1 + E_k) + \beta_k), \quad (10)$$

and the output is

$$y_k[n] = g_k a_k[n]. \quad (11)$$

A contextual alternative predicts all gates from the complete log-energy vector, $\mathbf{g} = \sigma(\text{MLP}(\log(1 + \mathbf{E})))$.

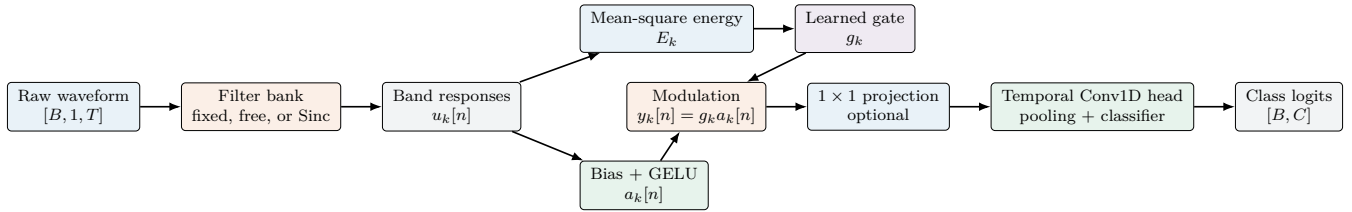


Figure 1: Complete Temporal EGFN flow. Unlike a standard Conv1D frontend, the model explicitly exposes band responses, mean-square energy, and multiplicative gates. The downstream temporal head retains ordering that the initial pooled classifier discarded.

3.3 Classifier evolution

The initial classifier pooled each band using mean, maximum, standard deviation, log-energy, and gate values before a small MLP. This representation was compact and interpretable, but global pooling removed temporal ordering. Temporal EGFN therefore sends the gated sequences through a lightweight Conv1D head. The wide configuration adds an $8 \rightarrow 32$ pointwise projection followed by 64- and 128-channel temporal blocks.

4 Hypotheses and Experimental Design

Five initial hypotheses guided the work:

H1: Frequency decomposition.

Band outputs should respond most strongly near their designed spectral regions.

H2: Energy-dependent modulation.

Gates should change as band energy changes.

H3: Predictive utility.

The frontend should support competitive audio classification.

H4: Observability.

Energy, gates, and filter responses should make internal behavior inspectable.

H5: Adaptation.

Learnable filters should improve difficult tasks, even if unconstrained learning weakens physical interpretation.

The experimental narrative follows the same sequence throughout: *problem* $\rightarrow$ *architectural change* $\rightarrow$ *hypothesis* $\rightarrow$ *experiment* $\rightarrow$ *visualization* $\rightarrow$ *revised question*. Models were implemented in PyTorch [8].

Accuracy is reported for balanced multiclass tasks. For MIMII, where normal recordings dominate, precision, recall, F1, and ROC-AUC are emphasized. Classification thresholds are selected on validation by maximizing F1 and are then frozen for test evaluation.

Table 1: Real-audio datasets and evaluated task.

Dataset	Task	Split focus
FSDD [6]	10 digits	random / speaker
Speech Commands [5]	10 keywords	official
MIMII [7]	valve anomaly	stratified

5 Synthetic Proof of Mechanism

5.1 Frequency response of the first prototype

The first model used five fixed filters and independent gates. A constant-RMS sinusoid was swept from 50 to 7950 Hz in 25 Hz increments and averaged over four phases.

Figure 2 supports H1: energy tracks the filter-bank structure. The gates range approximately from 0.533 to 0.677, supporting energy-dependent modulation but not discrete selection. Residual activation outside the nominal passband is expected from finite FIR responses, bias, and GELU.

5.2 Easy versus hard synthetic classes

The same checkpoint was evaluated without retraining on 2048 unseen easy and 2048 hard examples at 15 dB SNR. Easy classes occupied separated frequency regions; hard classes overlapped primarily between roughly 400 and 1500 Hz.

The failure in fig. 3 was informative: correct filtering does not guarantee sufficient class resolution. It motivated a finer eight-band bank and learnable filters.

5.3 Compact architecture progression

Table 2 summarizes the remaining synthetic iterations. Increasing frequency resolution and freeing the FIR coefficients improved the hard task, but free kernels became irregular and multi-band. Independent and contextual gates reached the same best validation accuracy. Sparsity regularization produced the highest value in this small experiment, although the gates remained soft rather than binary. Detailed filter sweeps and gate distributions are provided in the supplementary material.

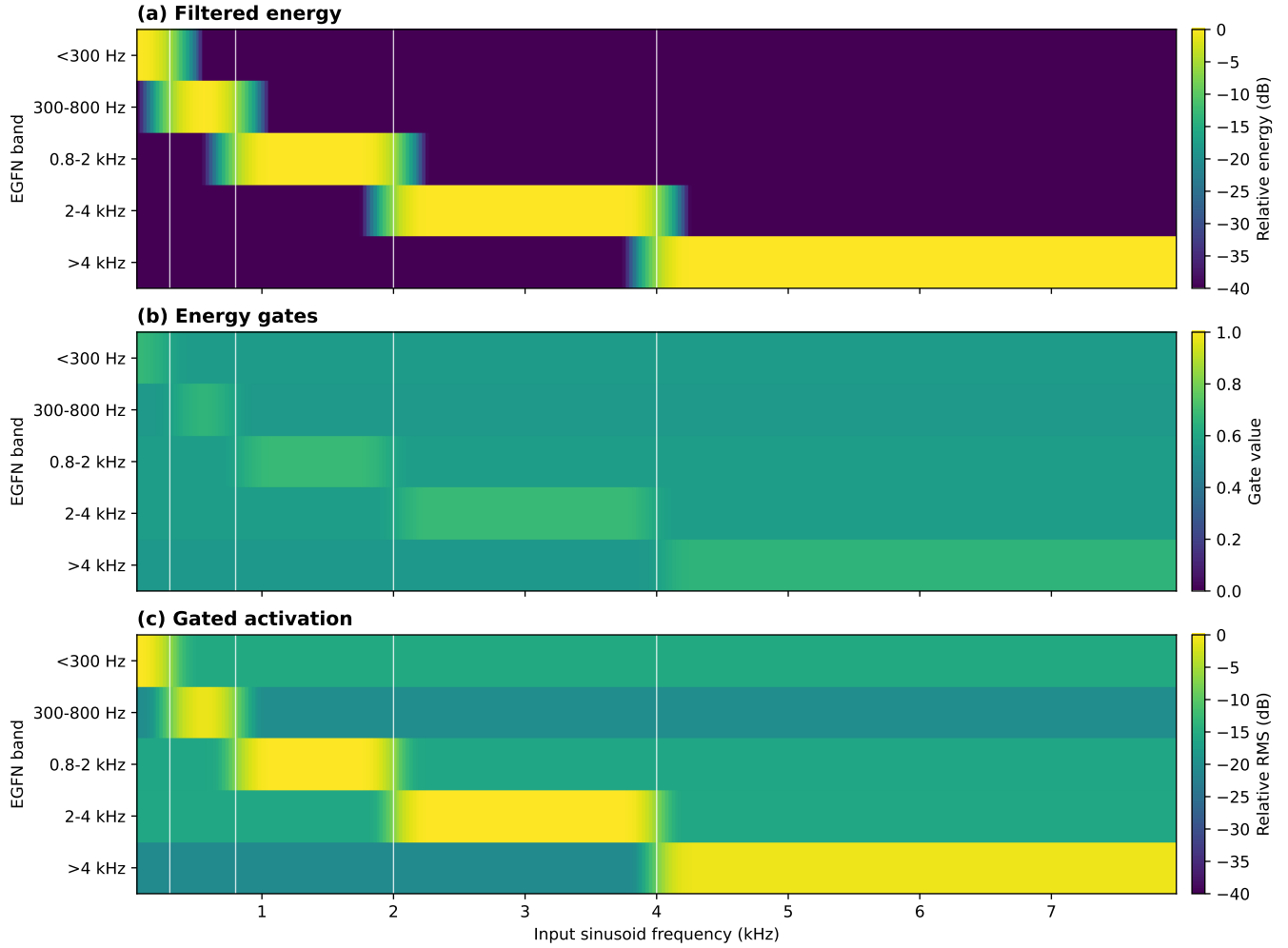


Figure 2: Sinusoidal activation sweep for the V0 checkpoint. Energy follows the expected coarse bands, while gates remain soft rather than binary. The figure validates frequency-dependent observability, not hard band selection.

Table 2: Synthetic hard-task progression. Results are best validation accuracies from the corresponding development runs.

Frontend	Gate configuration	Best val.
Five-band fixed	independent	0.531
Eight-band fixed	independent	0.633
Eight-band free FIR	independent	0.703
Eight-band free FIR	contextual	0.703
Eight-band free FIR	sparse penalty	0.727

These runs do not isolate the contribution of gating because every row retains an energy gate. Consequently, they establish architectural progression but not that gating outperforms an otherwise identical ungated filter bank.

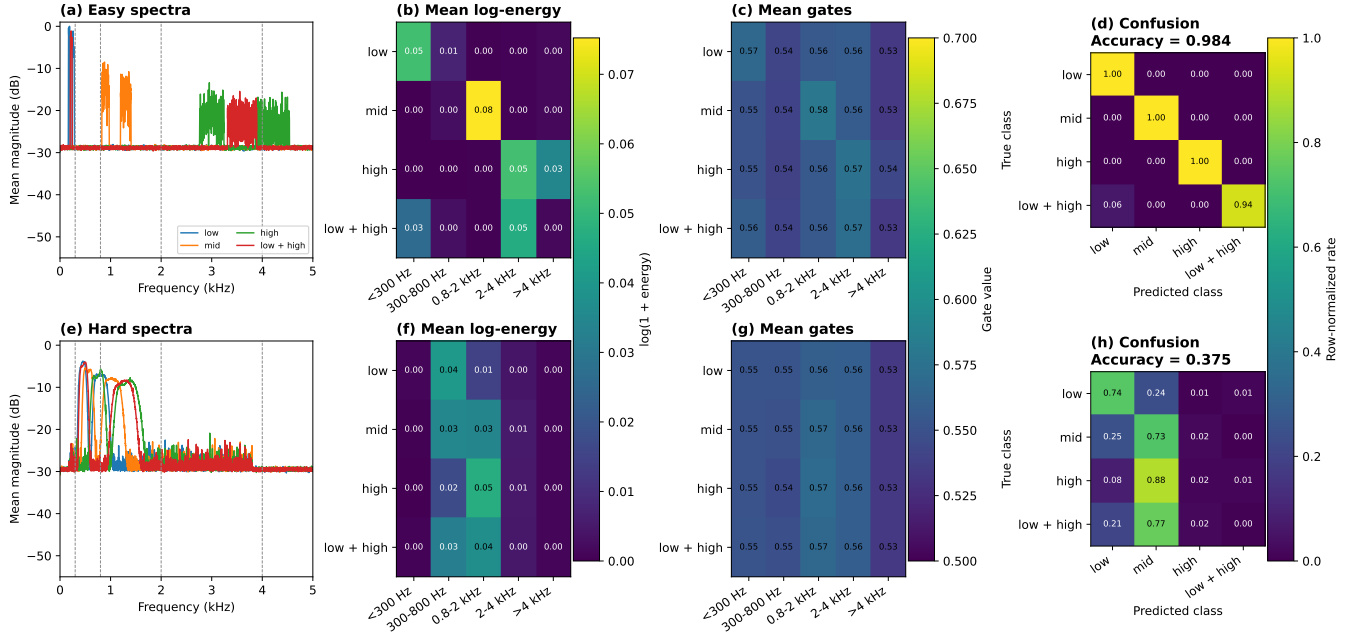


Figure 3: Behavior of one V0 checkpoint on easy and hard synthetic data. Test accuracy falls from 0.984 to 0.375 when class spectra overlap. The filters still respond, but the coarse representation no longer separates the labels.

6 From Synthetic Signals to Speech

6.1 FSDD: split and temporal structure

FSDD contains 3000 recordings of ten English digits from six speakers in the version used here [6]. A random recording split yielded test accuracy 0.840, but the same pooled EGFN fell to 0.560 when the test speaker was unseen. The random result is valid for its protocol, but cannot demonstrate speaker invariance. Replacing global pooling with a temporal head raised speaker-disjoint test accuracy to 0.708; Conv1D reached 0.746.

This stage demonstrates the importance of temporal structure, not an isolated advantage of the frequency front-end. Training histories and confusion matrices for both split protocols are retained in the supplementary material.

6.2 Scaling to Speech Commands

Speech Commands is designed for reproducible limited-vocabulary keyword spotting [5]. Ten commands were used with the official train/validation/test split: 30,769/3,703/4,074 recordings. Training included waveform augmentation.

Table 3: FSDD speaker-disjoint results.

Model	Params.	Best val.	Test
Pooled EGFN	4,106	0.538	0.560
Temporal EGFN	44,714	0.724	0.708
Conv1D	27,594	0.704	0.746

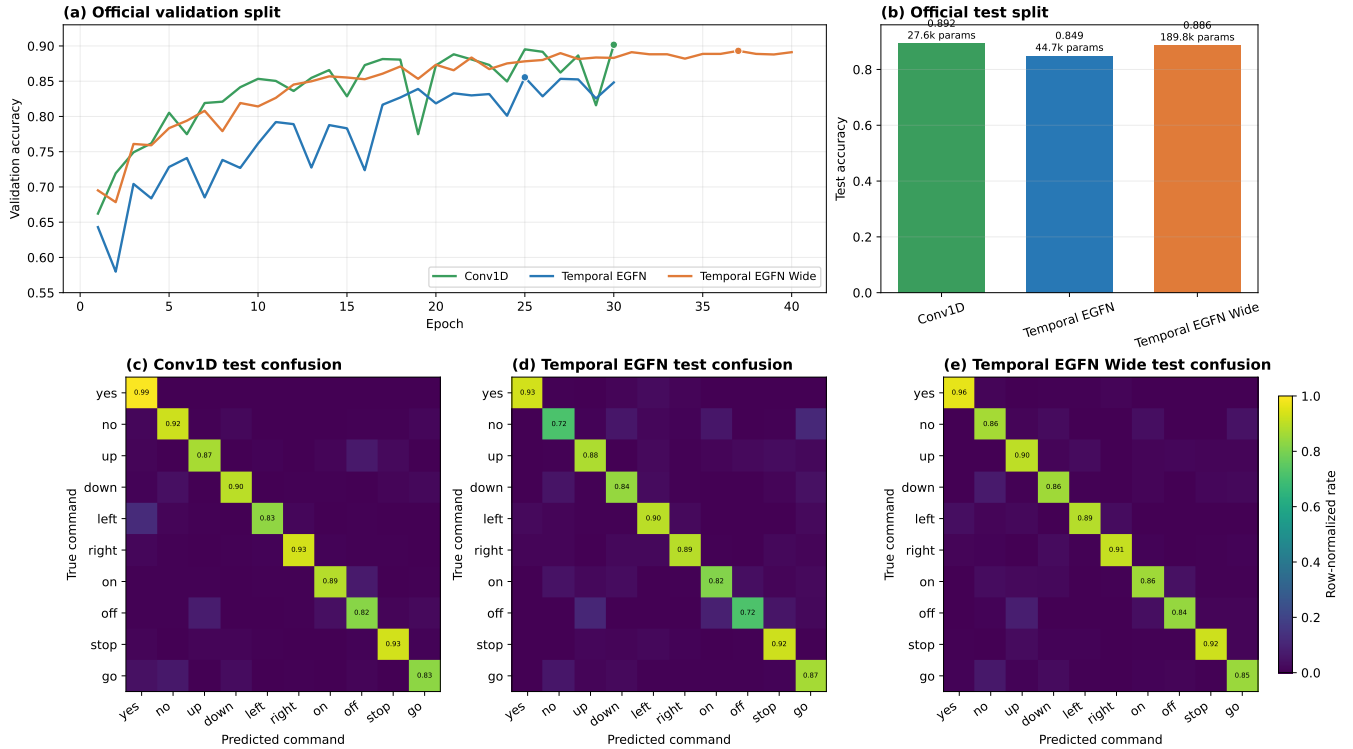


Figure 4: Speech Commands results. Temporal EGFN Wide reaches test accuracy 0.886, close to Conv1D at 0.892, and improves several class recalls. The near match requires substantially more parameters.

The wide model reduces the accuracy gap to 0.0066, showing that its frequency-gated representation can support competitive classification. However, it uses approximately 6.9 times the parameters of this Conv1D baseline. The result supports viability, not parameter efficiency.

Table 4: Speech Commands comparison.

Model	Params.	Best val.	Test
Conv1D	27,594	0.902	0.892
Temporal EGFN	44,714	0.856	0.849
Temporal EGFN Wide	189,802	0.893	0.886

7 Industrial Anomaly Detection

7.1 Why machinery is a relevant domain

MIMII contains normal and anomalous sounds from industrial machines, including valves, pumps, fans, and slide rails [7]. Mechanical faults can alter rotational components, harmonics, broadband friction, leakage, and high-frequency impacts. This makes machinery a stronger physical test of a frequency-structured frontend than spoken-word identity alone.

The valve subset used here contains 4170 recordings, split into 2922/624/624 train/validation/test examples. The test set contains 553 normal and 71 anomalous recordings. Balanced cross-entropy, augmentation, label smoothing, cosine scheduling, and early stopping were used.

7.2 Initial seed-42 application

The initial comparison used the original small Conv1D and Temporal EGFN Wide. Thresholds were selected on validation and applied once to test.

Calibration reduced EGFN false positives from 35 to 4, while recall decreased from 0.873 to 0.746. The strongest energy differences appeared above 4 kHz and in the initialized 1.5–2.5 kHz channel. Gate differences between normal and abnormal recordings remained small, with maximum magnitude near 0.012. Thus the model exposes useful internal measurements, but its gates are not class-specific switches.

7.3 Three-seed experiment

Seeds 42, 123, and 456 used the same protocol. Each run selected its threshold using validation only. With only three seeds, the mean and sample standard deviation below are descriptive summaries, not confidence intervals or evidence of statistical significance.

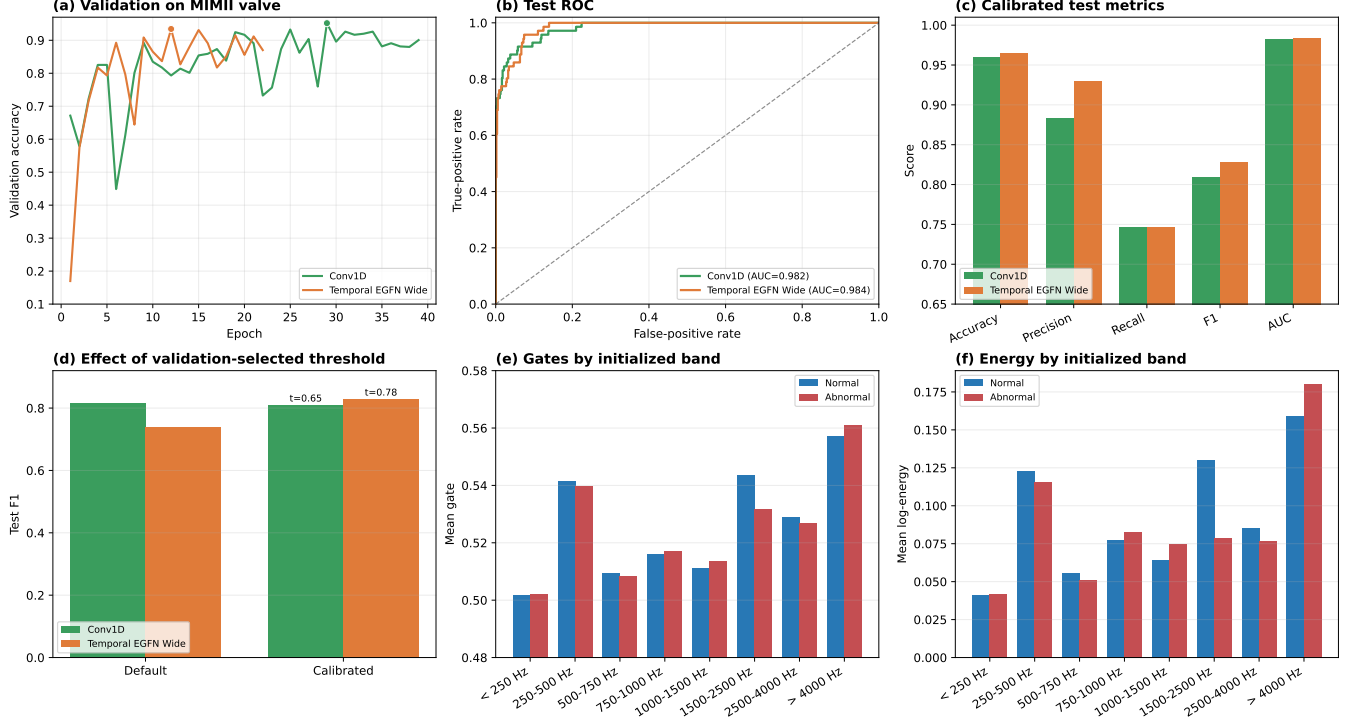


Figure 5: Initial MIMII Valve application. At validation-selected thresholds, EGFN reaches F1 0.828 and AUC 0.984, versus F1 0.809 and AUC 0.982 for Conv1D. Energy differences are more pronounced than gate differences.

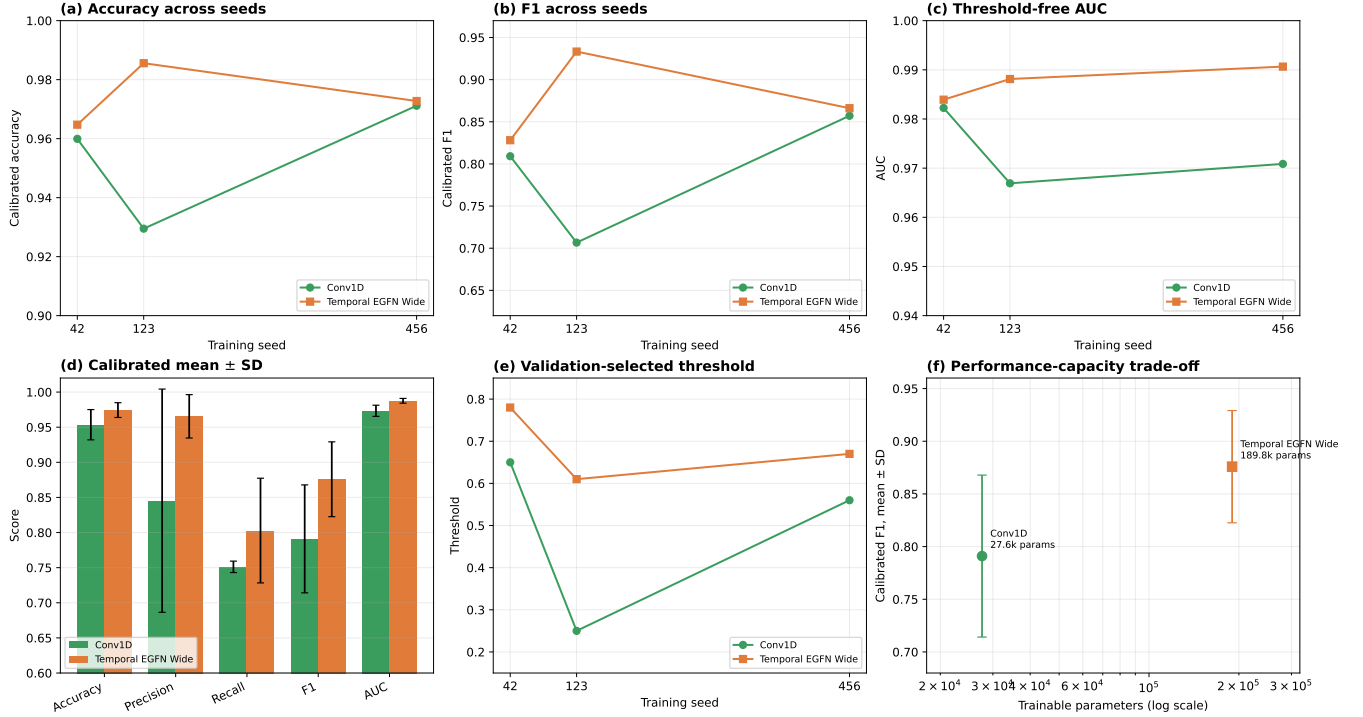


Figure 6: Preliminary three-seed MIMII comparison. Temporal EGFN Wide is higher in calibrated accuracy, F1, and AUC for these three runs. The performance–capacity panel also exposes the 6.9 $\times$ parameter gap.

The architectures are not capacity matched. The experiment therefore raises rather than resolves the attribution question: does EGFN improve because of its frontend or because its temporal model is larger?

8 Controlled V2 Frontend Study

8.1 Capacity-matched protocol

The final control fixes eight frontend channels, the $8 \rightarrow 32$ projection, the 64/128 temporal head, classifier, augmentation, loss, regularization, split, and seed. Only the frontend changes:

The free Conv1D and EGFN differ by only 16 parameters. In Sinc mode, sigmoid-parameterized lower cutoffs and positive bandwidths guarantee ordered cutoffs, at least 50 Hz bandwidth, and an upper cutoff below Nyquist. This control still compares only gated EGFN variants; it does not isolate gating from the filter bank itself. An ungated matched protocol is specified in the repository but is not reported because those runs were not yet completed.

8.2 Controlled result

Table 5: Descriptive MIMII mean $\pm$ sample standard deviation for three seeds; no significance claim is made.

Model	Accuracy	F1	AUC
Conv1D	0.954 ± 0.022	0.791 ± 0.077	0.973 ± 0.008
EGFN Wide	0.974 ± 0.011	0.876 ± 0.053	0.988 ± 0.003

Table 6: V2 controlled architectures.

Experiment	Frontend	Parameters
Matched Conv1D	free waveform conv.	188,754
EGFN-Free	free FIR + energy/gates	188,770
EGFN-Sinc	learned cutoffs + gates	187,978

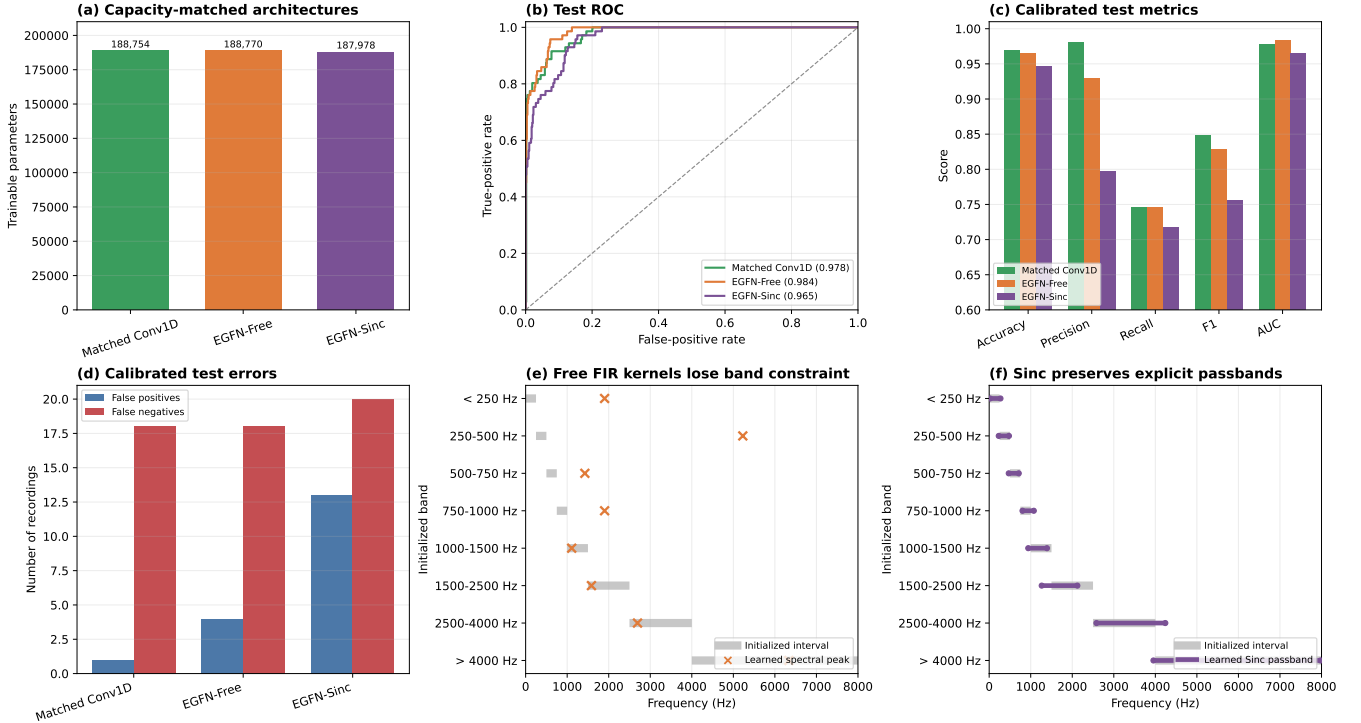


Figure 7: Capacity-matched V2 screening experiment on seed 42. Matched Conv1D obtains the best calibrated F1; EGFN-Free obtains the best AUC; EGFN-Sinc preserves explicit physical passbands but loses predictive performance.

The control changes the interpretation of V1. Once temporal capacity is matched, the unconstrained Conv1D frontend produces the best thresholded F1, while free EGFN retains a small AUC advantage. There is no uniform winner. The Sinc variant answers the technical criticism that free kernels can become ordinary convolutions: its filters remain explicit band-pass systems, but this constraint incurs a substantial performance cost in the seed-42 screening run. Because V2 contains one seed, it is evidence about this controlled run rather than a statistically definitive ordering.

9 Discussion

9.1 Answer to the central question

The experiments answer the original question in two parts. First, classical filtering principles can be embedded in a differentiable network, and explicit per-band energy can modulate neural activations. The sinusoidal sweep verifies the mechanism directly. It does not establish that learned gating improves prediction over the same filter bank with identity modulation. Second, the mechanism does not automatically produce better generalization or efficiency than Conv1D. Task difficulty, temporal modeling, filter freedom, model capacity, and threshold selection all matter.

Table 7: V2 seed-42 calibrated test results.

Model	Accuracy	F1	AUC
Matched Conv1D	0.970	0.848	0.978
EGFN-Free	0.965	0.828	0.984
EGFN-Sinc	0.947	0.756	0.965

9.2 Observability is not physical interpretation

EGFN provides *observability* by exposing filters, energy, and gates. Physical *interpretability* requires an additional link between those variables and a verified mechanism. In free mode, channel names cease to be physically guaranteed. In Sinc mode, cutoff frequencies remain meaningful, but a high gate still does not establish causal relevance. Per-band ablations and mechanical ground truth are needed to make that stronger claim.

9.3 Industrial diagnostic hypothesis

The MIMII results motivate, but do not demonstrate, diagnostic utility. The dataset labels used here distinguish normal from anomalous recordings; they do not connect each example to a verified fault mechanism or characteristic fault frequency. A future study could test whether energy or modulation changes align with rotational imbalance, misalignment harmonics, bearing frequencies, cavitation, leakage, friction, or wear when speed, load, and fault annotations are available. Until that experiment

is performed, industrial diagnosis is a hypothesis rather than a conclusion of this work.

9.4 Limitations

- The synthetic experiments simplify real acoustics and room response.
- FSDD has only six speakers; speaker-disjoint estimates are therefore sensitive to speaker selection.
- Speech Commands compares different parameter counts.
- The MIMII multiseed experiment uses only three seeds.
- V2 is a one-seed screening control and needs paired repetition.
- No matched filter-bank-without-gating ablation is reported, so the predictive contribution of the gate remains unidentified.
- SincNet and LEAF are discussed but not implemented as direct baselines.
- Mean gates are not a causal explanation and did not become hard sparse.
- Only MIMII valves were used, without sample-level physical fault labels.

9.5 Future work

The first follow-up is a paired ablation with an identical temporal head: free-filter bank without gating versus free-filter EGFN, and Sinc frontend without gating versus Sinc EGFN. Direct SincNet and LEAF baselines should then locate the architecture relative to established learnable frontends. These comparisons require paired multiseed repetition, latency and parameter counts, and threshold-free as well as operational metrics. Industrial validation should use datasets with machine type, speed, load, and verified fault mechanism so that learned bands can be compared against known physical frequencies.

10 Conclusion

Under capacity matching, Conv1D equaled or exceeded EGFN on the operational F1 metric; free-filter EGFN retained only a small ROC-AUC advantage in one seed-42 configuration. No general performance or efficiency advantage was established, and the contribution of gating itself remains unmeasured. The demonstrated contribution is instead a frequency-structured processing path in which band responses, energy, and learned modulation can be inspected. Free FIR filters were flexible but weakened physical meaning, while Sinc parameterization restored explicit passbands at a predictive cost in the controlled screening run. The project therefore contributes a tested architecture, a reproducible experimental progression, and a clear account of where the original hypothesis succeeded, failed, and remains open.

EGFN should presently be viewed not as a replacement for Conv1D, but as a research architecture for investigating frequency-dependent representation, adaptive modulation, and physically constrained filtering in neural audio models.

Its most compelling future role is in condition monitoring, where a model may be valuable not only because it detects a fault, but because its internal frequency variables can be compared with the physics of the machine.

Reproducibility Statement

All figures in this report are generated from saved checkpoints, histories, test diagnostics, and threshold reports in the project repository. Figure scripts are stored under `scripts/plot_*.py`. Data splits, seed values, parameter counts, and validation-selected thresholds are exported as CSV files. The project uses Python, PyTorch, torchaudio, NumPy, SciPy, scikit-learn, and Matplotlib. No test-set threshold optimization is used.

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